**Step 5: Test your logic and refine**

To assess the reliability of the automated feeding system, the system logic is verified in four simulated situations. The tested operations aim to ensure the effectiveness of the dispensing schedule, the accuracy of sensors, and the functionality of alert systems. Additionally, it identified areas for improvement based on false or malfunctioning steps.

The following are test scenarios and their outcomes.

1. **Pet Consumes Food as Anticipated**

- Input: The food container is adequately filled, and the scheduled feeding time arrives.
- Process: The servo motor operates, dispensing the designated portion; after 20 minutes, the weight sensor recognizes a notable reduction in the amount of food.
- Output: Feeding is completed, with no alerts generated.
- Result: The system functions well.

2. **Pet Does Not Eat**

- Input: The food container is adequately filled, and the scheduled feeding time is on; the food amount is unchanged after 20 minutes.
- Process: Weight sensor reading shows food amount ≥ threshold.
- Output: The system sends out an alarm (sound/light and direct message to staff).
- Result: Send alerts.
- Improvement: installing a camera to recognize whether the animal is present.

3. **Food Bin Empty Before Feeding**

- Input: Sensor identifies that the food level is below the predetermined < threshold before the scheduled feeding time.
- Process: the system ceases dispensing and sends staff alerts immediately.
- Output: Staff receives the alert; feeding is paused until food is refilled
- Result: System operates as expected
- Improvement: have a continuous alert indicator until the food is supplied.

4. **Manual Feeding Triggered**

- Input: The Staff operates a manual mechanism
- Process: The feeder dispenses food immediately.
- Output: feeding is logged; there are no conflicts with the schedule.
- Result: System functions correctly
- Improvement: Upgrade the logging process for manual feeding instances.

**Comparison with Expectations**

All simulated cases had outcomes that went along with the requirements set in the design from steps 1-4. The logic and sensors function promptly in terms of time to dispense, to send alerts, and to retain feeding.

**Suggested Improvements**

**The following are improvements identified from the simulated cases:**

- Install a small camera to trigger pet activities.
- Provide a continuous alert indicator when the food bin is empty, keep it active until the bin is refilled.
- Record all manual feedings and send alerts to track performance.
- Adjust waiting time after feeding based on animal behavior, with the setting configured by staff.